Section 3.7- Foundation Models for Genomics

2026 Spring

LLM Agents Foundation & Applications

Dr. Yanjun Qi

20260210

Road Map (Five papers · 2025–2026)

1	Genomic Language Models at Scale	Evo 2 (Brixi et al., 2025)
1	Architecture: Beyond Transformers	StripedHyena 2 (Ku et al., 2025)
3	Regulatory Genome Interpretation	AlphaGenome (Avsec et al., 2026)
4	Single-Cell Foundation Models	scGPT (Cui et al., 2025)
5	Benchmarking Virtual Cells	Virtual Cell Challenge (Roohani et al., 2025)
6	Synthesis & Cross-Cutting Themes	Discussion

Creation of this slide deck was assisted by GenAI tools.

The Big Picture: Three Layers of Genomic AI

Layer 1 · Sequence Architecture – Long context for genomic sequence

How do we efficiently model DNA at genome scale (1M+ bp) with base-pair resolution? What operators handle long-range dependencies without prohibitive memory cost?

StripedHyena 2 - Evo 2 ·

Layer 2 · Sequence-to-Function

Can a single model trained on billions of nucleotides learn biology from raw sequence? Predict mutation effects, generate functional genomes, interpret regulatory elements.

AlphaGenome

Layer 3 · Cell-State Modeling

How do cells respond to perturbations? Predict post-perturbation transcriptomes, model causal intervention effects, benchmark models against real biology.

scGPT · Virtual Cell Challenge

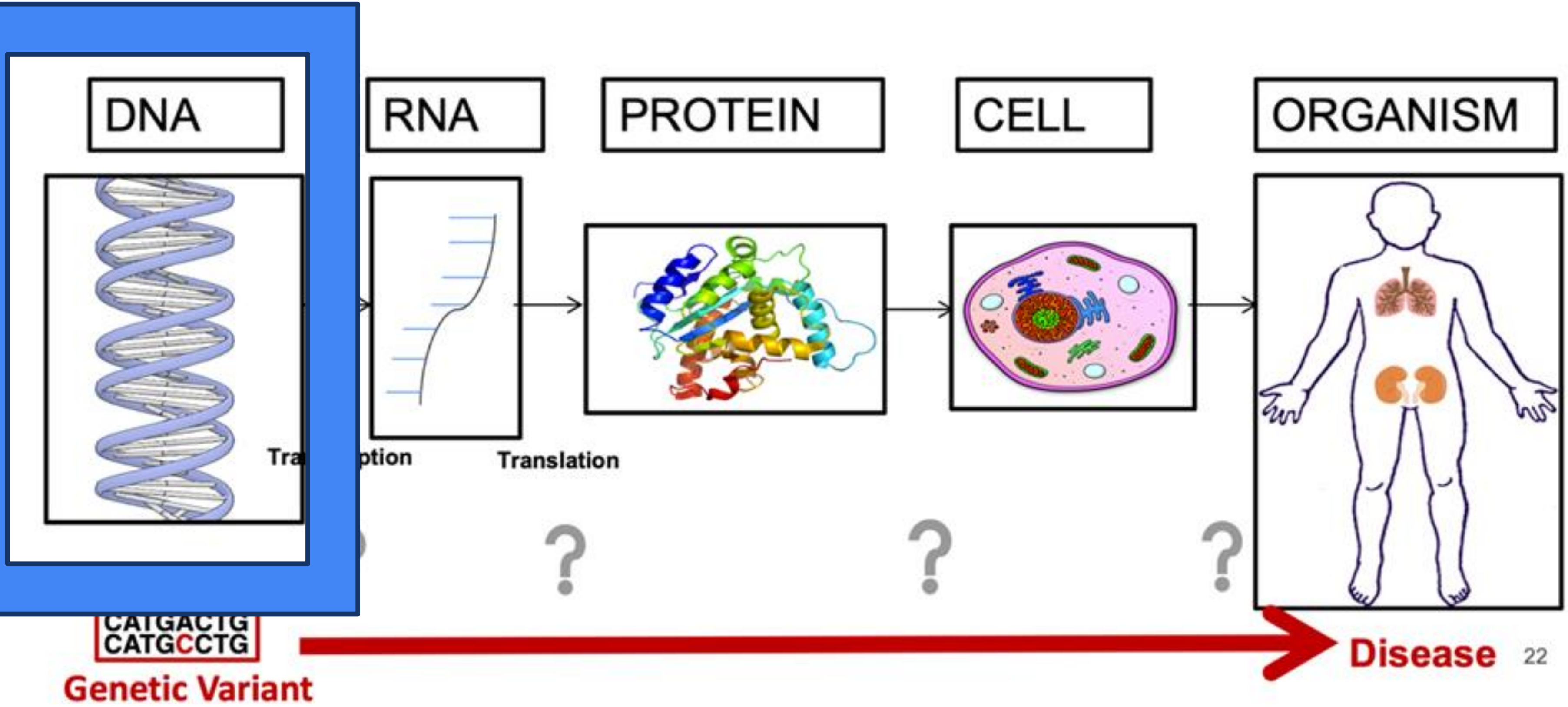
1

Evo 2

Genome Modeling and Design Across All Domains of Life

Brixi et al. · Arc Institute / Stanford / UC Berkeley / NVIDIA · bioRxiv 2025

Biology in a Slide:



Motivation: A Generalist Genomic Language Model

The Gap in Genomic Understanding

All life is encoded in DNA — yet we cannot 'read' genomes the way we read sentences. Most ML models are narrow:

- Protein-only (ESM, AlphaMissense)
- Prokaryote-only (Evo 1, Nucleotide Transformer)
- Coding-region-only (miss 98%+ of human variation)
- No alignment-free model can score indels + splicing + pathogenicity simultaneously

Key Hypothesis: Training a sufficiently large language model on diverse genomic sequences spanning all domains of life will yield emergent understanding of biological function — without task-specific labels.

Evo 2: Scale Comparison

	Evo 1	Evo 2
Parameters	7B	40B
Tokens	300B	9.3T
Context	8K	1M
Domains	Prokaryotes	All life
Modalities	Limited	Coding + noncoding + splicing + generation

Evo 2

Training Data: OpenGenome2 & Two-Phase Strategy

OpenGenome2: 8.84 Trillion Nucleotides

Source	Size
Eukaryotic genomes (15,032 species)	6.98T nt
Prokaryotic genomes (GTDB)	357B nt
Metagenomes	854B nt
Organelle genomes (32K)	2.82B nt
mRNA / ncRNA transcripts	~600B nt

Single-nucleotide (byte) tokenization — no vocabulary needed. Eukaryotic virus sequences excluded for biosafety.

Two-Phase Training Strategy

Phase 1 · Pretraining

8,192 tokens context

Weighted toward coding regions and functional elements. 8.7T tokens for Evo 2 40B. Learns sequence-to-function mapping.

Phase 2 · Midtraining

Context extension to 1M

Gradually extends: 32K → 65K → 131K → 262K → 524K → 1M. Rotary embedding base frequency scaled 10× per doubling. More whole-genome sequences.

Needle-in-a-haystack test confirms successful retrieval of information at all positions within 1M-token context after midtraining.

Use: Zero-Shot Mutational Effect Prediction

Zero-shot scoring: Δ log p(mutant) − log p(reference). A mutation that decreases sequence likelihood → predicted to be deleterious. No fine-tuning or labels required.

Codon Structure (36 species)

Strong likelihood drops at start codons. 3-nucleotide periodicity in CDS (wobble positions spared). Ribosome binding sites (prokaryotes) and Kozak consensus (eukaryotes) detected — without annotation.

Mutation Sensitivity Hierarchy

Frameshift > Premature stop > Non-synonymous > Synonymous in coding sequences.

tRNA/rRNA deletions >> intergenic deletions in noncoding regions.

All learned from raw sequence, no labels.

DMS & Essentiality

Competitive with state-of-the-art protein LMs on bacterial & human proteins.

State-of-the-art on ncRNA fitness prediction.

lncRNA essentiality in humans: substantially outperforms Nucleotide Transformer.

Evo 2

Use: Clinical Variant Effect Prediction

Benchmark Suite

Benchmark	N variants	Evo 2 strength
ClinVar coding SNV	14,319	Ranked 4th–5th
ClinVar noncoding SNV	34,761	Best performance
ClinVar non-SNV (indels)	5,130	Best — others can't score indels
SpliceVarDB	4,950	Best zero-shot
BRCA1 saturation mutagenesis	3,202	Best overall
BRCA2	6,837	Best overall

BRCA1 Supervised Classifier

- Extract Evo 2 40B embeddings from layer 20
- Train tiny feedforward: Linear(512) → Linear(128) → Linear(32) → Sigmoid
- 80/20 train/test split on BRCA1 SNVs

0.94 AUROC — coding SNVs

0.95 AUROC — all SNVs

Evo 2 uniquely handles indels, deletions, and noncoding variants — classes where protein-specific models (AlphaMissense, GPN-MSA) cannot produce scores at all.

Genome-Scale Generation & Generative Epigenomics

What Can Evo 2 Generate?

Human Mitochondrial Genome (~16 kb)

Correct counts: 13 CDS, 22 tRNA, 2 rRNA genes. Correct synteny. 76–92% identity to human proteins. AlphaFold 3: expected mitochondrial complex structures (CI–CV).

M. genitalium genome (~580 kb)

~70% of predicted genes have Pfam hits (vs. 18% for Evo 1). Protein secondary structure distributions match natural genome.

S. cerevisiae chr III (~330 kb)

Contains tRNAs, promoters, and genes with intronic structure. Protein homology to natural yeast genes confirmed.

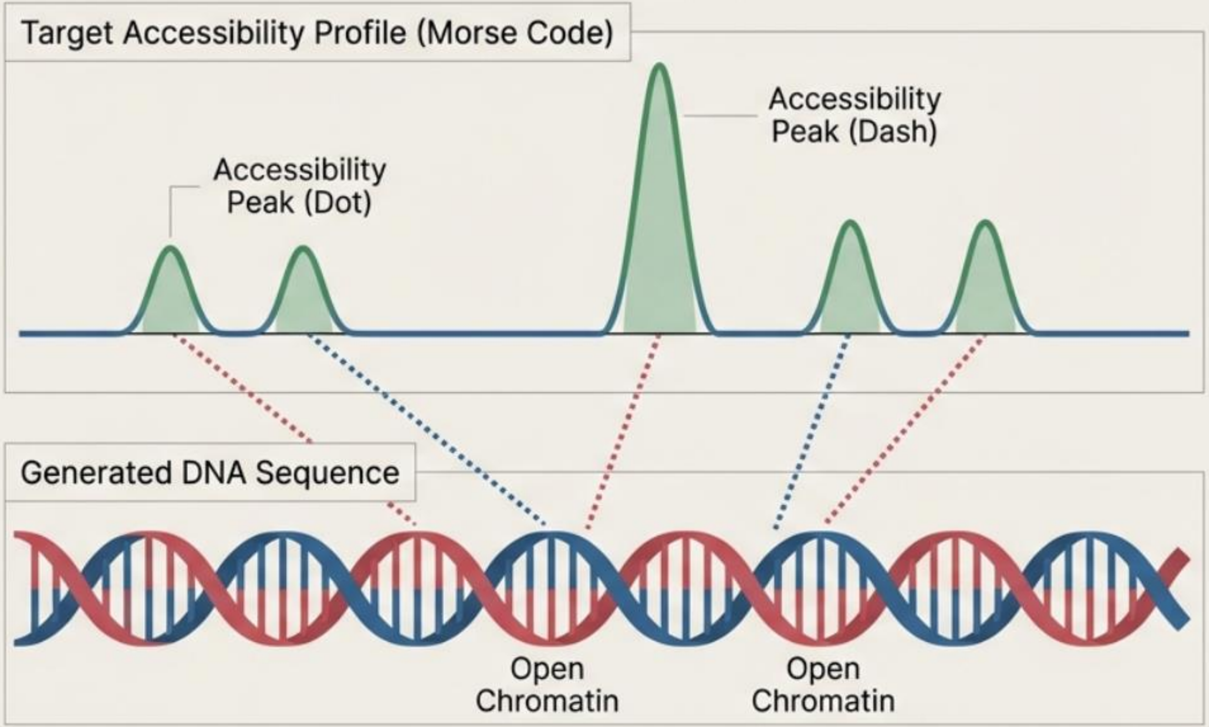
Generative Epigenomics via Inference-Time Search

- Goal: Design DNA with specified chromatin accessibility profiles (open vs. closed regions)
- Approach: Beam search + scoring. Evo 2 proposes 128-bp chunks; Enformer + Borzoi score chromatin accessibility; keep best-matching chunks; repeat autoregressively.
- Key finding: Increasing inference-time compute (tokens sampled per bp) predictably improves design quality in a log-linear relationship.

AUROC ≈ 0.9 at 30+ tok/bp across diverse peak patterns. First demonstration of inference-time scaling in biological language modeling. Generated sequences maintain natural dinucleotide frequencies.

"Morse code" messages (LO, ARC, EVO2) written into designed epigenomes as a proof-of-concept of programmable chromatin pattern design.

Generative Epigenomics via Inference-Time Search



Inference-Time Search allows us to guide generation to satisfy specific constraints.

Demo: Successfully encoded Morse code into the physical chromatin accessibility of a DNA DNA sequence.

1.5

StripedHyena 2

Systems & Algorithms for Convolutional Multi-Hybrid Language Models at Scale

Ku, Nguyen, Romero et al. · Stanford / Arc Institute / NVIDIA · arXiv 2025

Why Go Beyond Transformers?

The Transformer Bottleneck

- Attention is quadratic in sequence length — at 1M tokens, full attention is prohibitively expensive.
- Existing hybrids (SSMs, linear attention) only beat Transformers at very long sequences — exactly where they underperform on quality.
- Previous architecture alternatives fail to deliver consistent gains at typical pretraining regime (short sequences, wide models).

Key Insight: The recall specialization problem

Most hybrid operators compete for the same capability — in-context recall — leading to redundancy. Different operators actually excel at different subtasks.

Operator Specialization

Operator	Strength
Short FIR convolution	Local context, tensor core utilization
Input-dep. convolution	Multi-token recall, noise filtering
Full self-attention	Long-range targeted recall
Hyena-LI (implicit)	Full sequence context, constant memory

Architecture: Three Complementary Operators

Hyena-SE (Short Explicit)

Range / Filter

Local patterns
Filter length: 7

Key strength
Maximizes hardware utilization via tensor cores. Specializes in local multi-token recall and byte-level pattern matching.

Hyena-MR (Medium Regularized)

Range / Filter

Medium range
Filter length: 128

Key strength
Analogous to sliding window attention. Exponential decay regularizer forces focus on recent context.

Hyena-LI (Long Implicit)

Range / Filter

Full sequence
Implicit filter

Key strength
Real exponential basis captures arbitrary long-range dependencies. Supports recurrent parametrization for constant memory at inference.

StripedHyena 2 interleaves SE → MR → LI in a striped pattern, with 5 sparse attention (MHA) layers throughout. Filter grouping turns GEMV → GEMM, enabling full tensor-core utilization.

StripedHyena 2

Hardware-Aware Algorithms: Blocked Convolution

The Problem with Standard Libraries

PyTorch and cuDNN handle short CNNs (Winograd) and very long FFT convolutions well — but are not optimized for grouped depthwise convolutions of varied lengths used in multi-hybrids.

The Two-Stage Blocked Kernel Solution

Partition input/output into chunks of size ℓ_b. The Toeplitz convolution matrix decomposes into just two block types:

- H_0 — current chunk (block-diagonal GEMM)
- H_1 — previous chunk spillover (off-diagonal GEMM)
- Each output block = $H_0 \times v\_n + H_1 \times v\_\{n-1\}$ → two GEMMs per block, H_0/H_1 loaded once, reused across all channels in the group.

Context Parallelism for 1M-Token Training

Strategy	Best For	How
All-to-All	Hyena-LI, MHA	Each rank exchanges data; processes full sequence in hidden-dim shards
Point-to-Point	Hyena-SE, MR	Only ℓ_h − 1 boundary elements exchanged with neighbor rank

Filter Grouping: Filters shared across groups of d_g channels → transforms GEMV to GEMM → full tensor core utilization on H100 GPUs. No significant quality impact (ablated: group size 1 vs 16).

StripedHyena 2

Results: Speed, Quality, and Scale

End-to-End Training Throughput

1.2 – 2.9×

faster than optimized Transformers (7B & 40B scale)

1.1 – 1.4×

faster than StripedHyena generation 1

2×

operator throughput vs. linear attention / SSMs (D=4096, H100)

SE-MR-LI layout achieves the best perplexity among all multi-hybrid configurations on OpenGenome2 (7B, 400B tokens). Successfully extended to 1M context length via midtraining with PI+ABF position interpolation.

Model Variants

Model	Params	Tokens	Context
Evo 2 40B	40.3B	9.3T	1M
Evo 2 7B	6.5B	2.4T	1M
Evo 2 1B	1.1B	1T	8K

Validated at 40B parameters on OpenGenome2 — one of the largest fully open AI models across any modality.

Fully open: model weights, training code (Savanna framework), and OpenGenome2 dataset released publicly at github.com/Zymrael/savanna

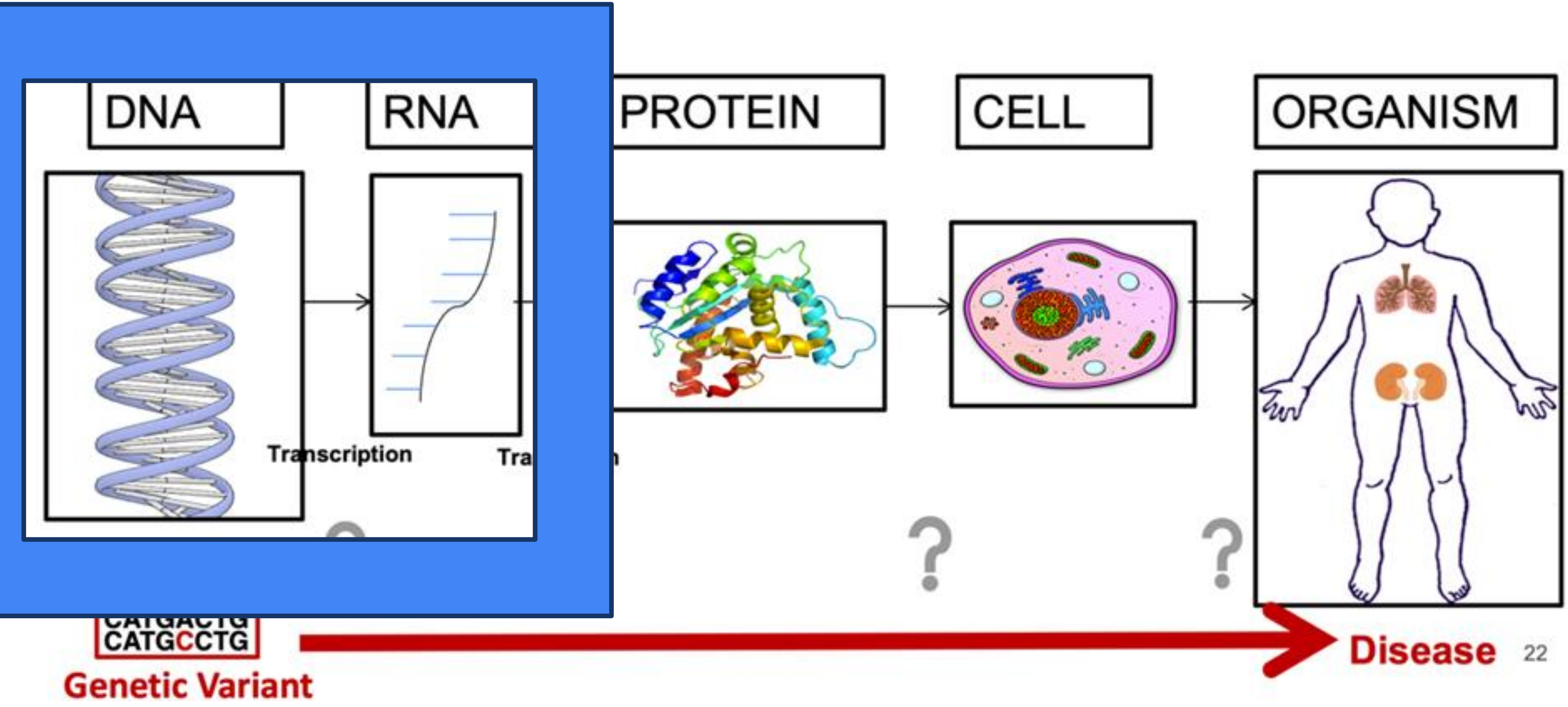
3

AlphaGenome

Advancing Regulatory Variant Effect Prediction

Avsec et al. · Google DeepMind · Nature, Vol 649, January 2026

Biology in a Slide:



AlphaGenome

The Challenge: Non-Coding Variant Interpretation

98%+ of human genetic variation is non-coding — yet we have limited tools to interpret it.

Non-coding variants can affect:

- Chromatin accessibility and epigenetic marks
- 3D chromatin conformation (enhancer–gene looping)
- Gene expression levels across tissues
- Splicing patterns and isoform composition

AlphaGenome's Innovation

1 Mb DNA input → 5,930 human + 1,128 mouse genome tracks → single base-pair resolution across 11 modalities

First model to simultaneously achieve: long context + bp resolution + all modalities

Prior Models and Their Trade-offs

Trade-off	Short models	Long models
Resolution	Base-pair (SpliceAI, BPNet)	Coarse bins (Enformer: 128bp)
Context	≤10 kb — miss distal enhancers	200–500 kb
Modalities	Specialized (1–2)	Multi-modal

11 Predicted Modalities:

- RNA-seq, CAGE, PRO-cap (gene expression)
- DNase-seq, ATAC-seq (chromatin accessibility)
- Histone ChIP-seq, TF ChIP-seq
- Splice sites, splice site usage, splice junctions
- DNA contact maps (Hi-C, Micro-C)

AlphaGenome Architecture: U-Net + Transformer

Three-Stage U-Net Backbone

Encoder
Convolutional blocks progressively downsample 1 bp → 128 bp resolution, increasing feature channels. Captures local sequence features.

Transformer Tower
Operates at 128 bp resolution. Models long-range dependencies (enhancer–promoter interactions) via attention across 1 Mb context.

Decoder
Upsamples back to 1 bp via conv blocks + skip connections from encoder. Preserves spatial detail for base-pair resolution output.

Two representation types: 1D embeddings (1 bp & 128 bp) → linear genome predictions; 2D embeddings (2,048 bp) → pairwise chromatin contact maps.

Enabling 1 Mb at Base-Pair Resolution

The Challenge
Processing 1 Mb at 1 bp resolution requires enormous memory. Naive approach is infeasible.

Solution: Sequence Parallelism
1 Mb sequence split into 131 kb chunks across 8 interconnected TPU v3 devices. Transformers communicate between devices to share long-range context.

Why 1 Mb?
99% of validated enhancer–gene pairs (465/471) fall within 1 Mb. Context is designed to encompass the full relevant distal regulatory landscape.

Novel Splice Junction Head: First model to jointly predict splice sites, usage, and junction counts. Models donor–acceptor pair interactions rather than predicting each position independently.

AlphaGenome

Training: Pretraining + Knowledge Distillation

Stage 1 · Pretraining

- Trained on human and mouse genomes
- 4-fold cross-validation scheme: ¾ of reference genome for training, ¼ held out
- Produces fold-specific models (for unbiased evaluation) and all-fold models (trained on all data)
- Data augmentation: random shift + 50% reverse complement

Stage 2 · Distillation

- Ensemble of all-fold models → teachers (up to 64 models)
- Single student model trained to reproduce teacher outputs
- Student trained on randomly augmented AND mutationally perturbed sequences
- Speed: <1 second per variant on NVIDIA H100 GPU

Why Distillation?
• Single inference pass per variant → fast, scalable screening
• Perturbation during distillation improves robustness to sequence variation
• Single distilled model competitive with 4-model pretrained ensemble

Benchmark scope:
• 24 genome track prediction tasks (11 modalities)
• 26 variant effect prediction tasks
• vs. Enformer, Borzoi, SpliceAI, Pangolin, ChromBPNet, Orca, ProCapNet

AlphaGenome

Results: Benchmark Performance

22/24	25/26	+41%	+42%
genome track prediction tasks won	variant effect prediction tasks won	more eQTLs recovered vs. Borzoi at 90% sign accuracy	contact map cell-type differences vs. Orca

Selected Improvements over Best External Model

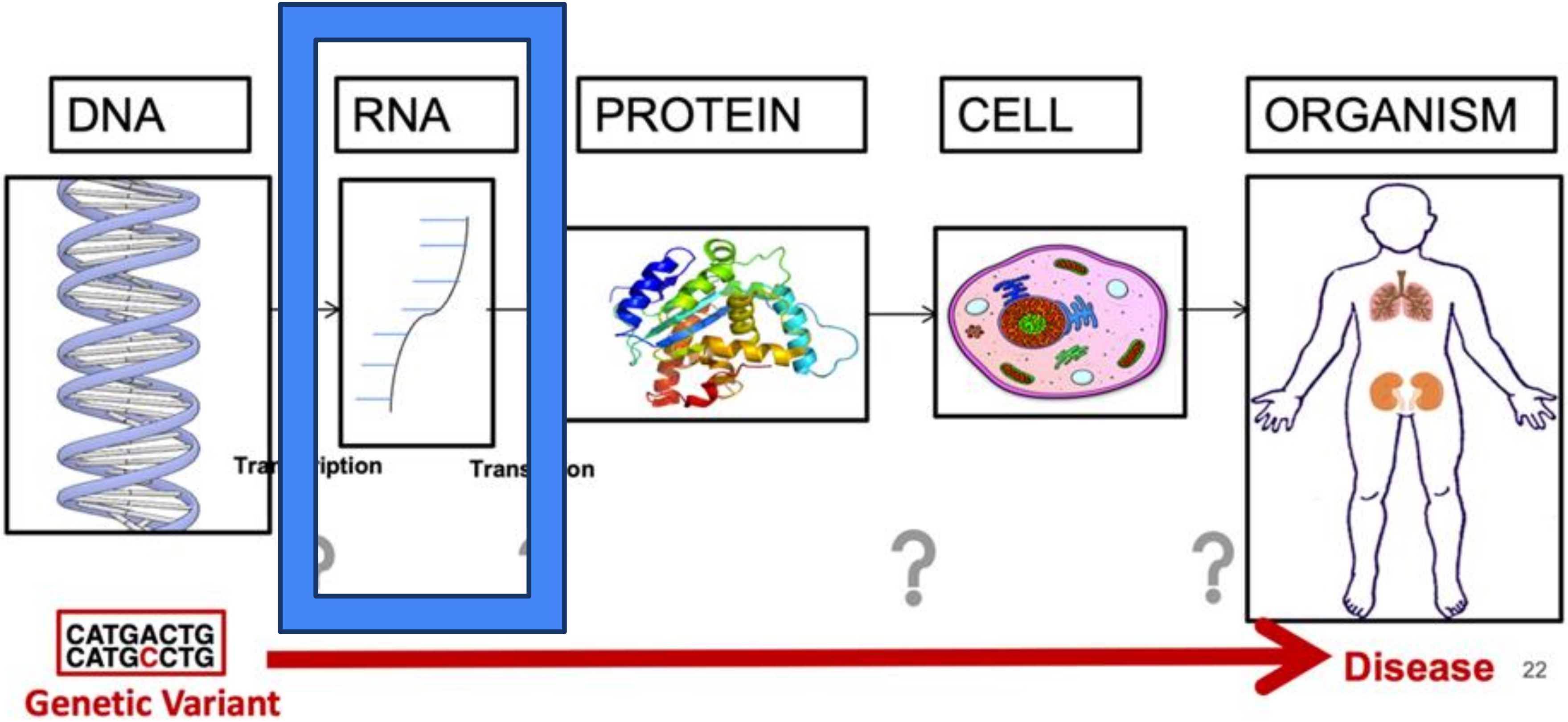
Modality / Task	AlphaGenome	Best Competitor	Improvement
eQTL sign prediction (SNVs)	auROC 0.80	Borzoi 0.75	+6.7%
eQTL coefficient (Spearman ρ)	0.49	Borzoi 0.39	+25.6%
Splicing outlier (zero-shot)	auPRC 0.22	Pangolin 0.18	+22% rel.
sQTL causality (≤10kb)	auPRC 0.76	Pangolin 0.73	+4.1%
Contact map cell-type diff.	Pearson r 0.63	Orca	+42.3%
CAGI5 MPRA (multimodal)	Pearson r 0.65	Borzoi ens. 0.63	New SOTA

4

scGPT

Large Models for Single-Cell Omics & Drug Discovery
Haotian Cui · GenBio AI · scGPT · LUMI-Lab · State-Conditioned Perturbation Modeling

Biology in a Slide:



What Is Single-Cell RNA Sequencing?

Traditional RNA-seq → bulk average across millions of cells

scRNA-seq → gene expression of each individual cell

Analogy

Bulk RNA-seq is like blending a fruit salad and tasting the mix. scRNA-seq is tasting each fruit separately — you learn what makes each one unique.

Cell Heterogeneity

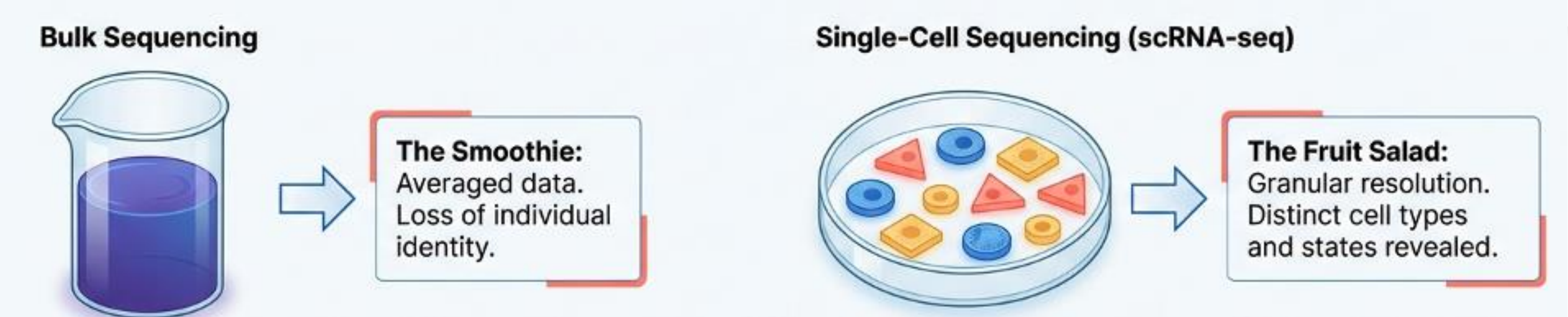
Reveals cell-to-cell differences hidden in bulk data

Spatial Transcriptomics

Maps gene expression to tissue locations

Massive Scale

Thousands to millions of cells per experiment



The Traditional scRNA-seq Pipeline

Every analysis follows a similar chain of steps — each requiring tool choices and parameter tuning.

1
QC & Filtering
Remove low-quality cells & doublets

→

2
Normali-zation
Log-normalize, scale gene counts

→

3
Feature Selection
Pick highly variable genes (HVGs)

→

4
Dim. Reduction
PCA → UMAP / t-SNE embedding

→

5
Clustering
Leiden / Louvain community detection

→

6
Annotation
Label cell types from marker genes

Downstream Tasks

Batch Correction

Trajectory Inference

Spatial Domains

Gene Imputation

DEG Analysis

⚠ **Each step needs tool selection + parameter tuning + biology expertise + Python coding**

Motivation: The Single-Cell Scale Opportunity

Single-cell RNA sequencing now produces tens of millions of profiled cells across labs, diseases, and species. This scale makes pretraining viable — and reveals biological structure invisible in small studies.

What Foundation Models Offer:

- Pretrain on millions of cells across diverse studies
- Learn gene–gene and cell–state structure at scale
- Transfer to annotation, mapping, perturbation prediction with few-shot adaptation

Key principle: Always compare against GEARS and CPA baselines. Any model that cannot beat these is not useful.

scGPT Architecture & Input Representation

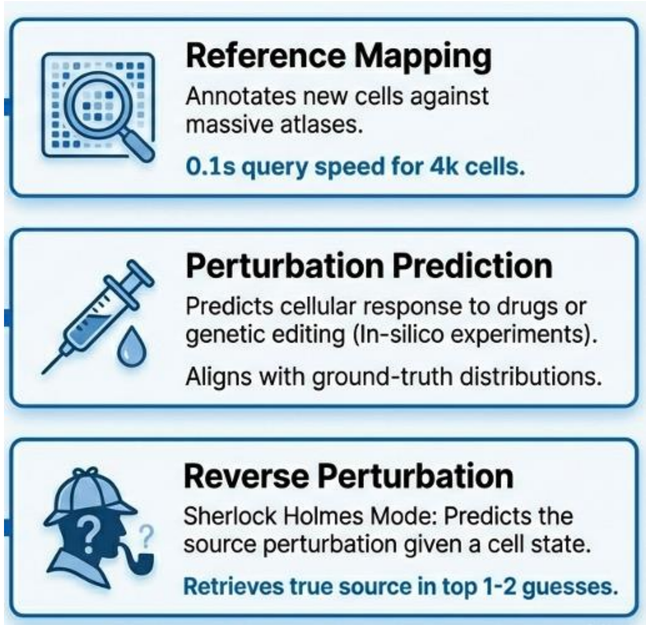
Transformer Encoder for Single Cells

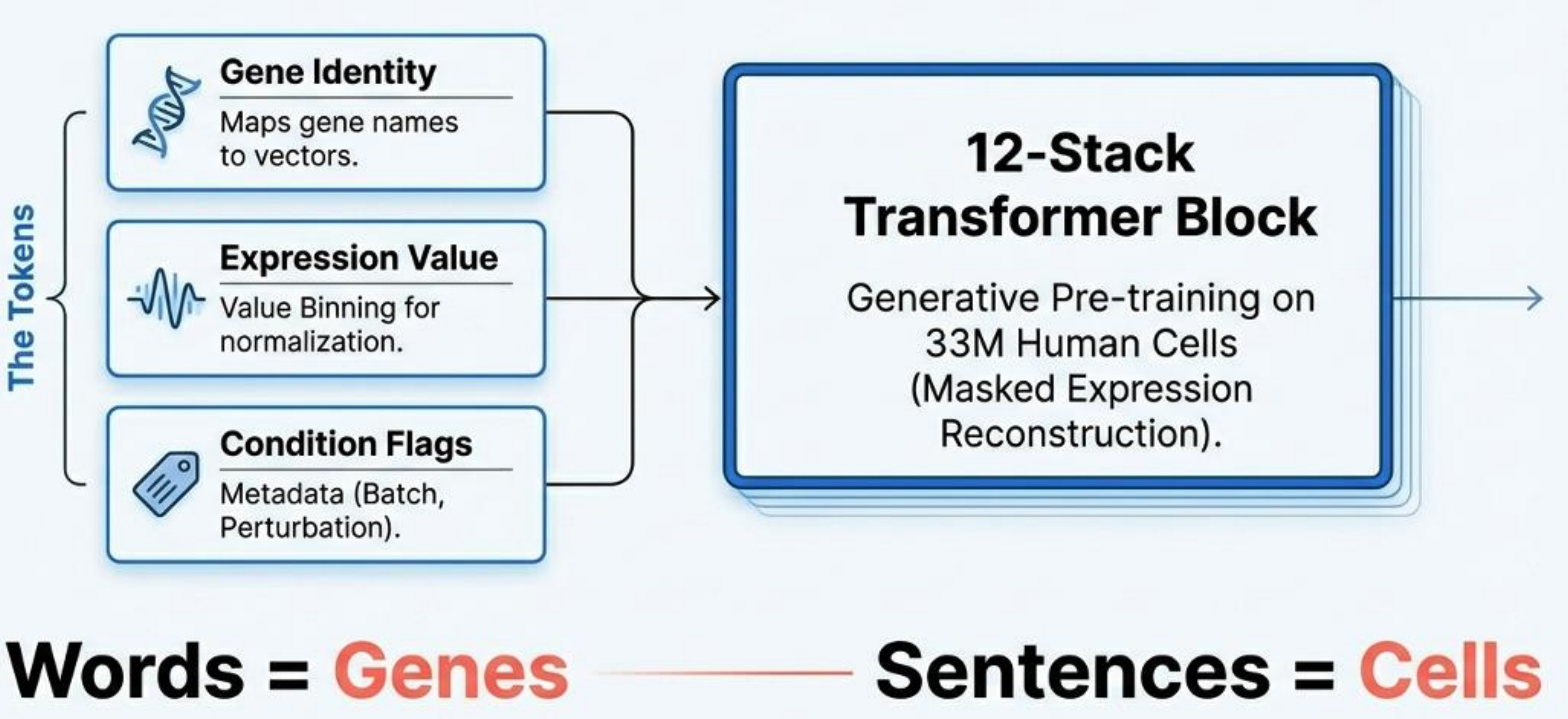
Core idea: A transformer encoder trained with masked gene expression modeling over a massive single-cell corpus, yielding a general-purpose cell embedding via a CLS token.

Component	Details
Input tokens	(gene_id, binned_value) pairs
Value encoding	Rank or value binning — stabilizes cross-study variation
Optional covariates	Batch ID, tissue, species
Attention	Gene-level multi-head self-attention
CLS token	Aggregates whole-cell state for downstream tasks
Pretraining	Masked gene modeling — predict held-out gene values

State-Conditioned Perturbation Modeling

Separate state from effect. First encode the baseline cell. Then encode the intervention. Model their interaction to predict the post-perturbation state.





The Tokens
Gene Identity
Maps gene names to vectors.
Expression Value
Value Binning for normalization.
Condition Flags
Metadata (Batch, Perturbation).
12-Stack Transformer Block
Generative Pre-training on 33M Human Cells (Masked Expression Reconstruction).
Words = Genes
Sentences = Cells

Pre-training Strategy

Data

- 10.3 million human PBMC / bone marrow cells
- 65 datasets from CellXGene portal
- Filtered: Homo sapiens, blood/bone marrow, normal + COVID-19/influenza conditions

Generative Pre-training Objectives

GEP — Gene Expression Prediction

Randomly mask a subset of gene expression values; predict them from the unmasked context using cross-entropy loss.

GEPC — GEP for Cell Modelling

Same as GEP, but predictions are made from the CLS cell embedding — directly optimizes cell representations. Combining GEP + GEPC significantly outperforms either alone.

10.3M

single cells pre-trained on

65

datasets from CellXGene

6

training epochs · Adam · lr = 1×10^{-4}

Fine-tuning Objectives

Abbreviation	Full Name	What It Does
GEP	Gene Expression Prediction	Reconstruct masked gene values from context
GEPC	GEP for Cell Modelling	Reconstruct gene values from CLS cell embedding
ECS	Elastic Cell Similarity	Contrastive: push similar cells closer, dissimilar apart
DAR	Domain Adaptation via Reverse Backprop	Gradient reversal to remove batch identity from embeddings
CLS	Cell Type Classification	Cross-entropy loss against ground-truth cell type labels
perturb-GEP	Perturbation GEP	Predict post-perturbation expression from control cell

Different downstream tasks combine different subsets of these objectives. All inherit the pre-trained weights as initialization.

Training Objectives & Benchmarking Framework

Four Core Training Objectives

Objective	Input	Target	Phase
Masked Gene Modeling	Partial expression profile	Held-out gene values	Pretraining
State Transition	Expression at time t	Expression at t + Δ	Fine-tuning / PT
Perturbation Prediction	State + edit	Post-perturbation Δ	Fine-tuning
Reverse Identification	Δexpression	Applied perturbation	Fine-tuning

Design principle: Each downstream task has an aligned training signal — not just a reconstruction proxy. Misalignment between pretraining loss and task utility is treated as a first-class problem.

Honest Benchmarking Requirements

- Baselines: always compare GEARS and CPA — any model below these is not useful
- Splits: held-out cell types AND held-out perturbations; report task-aligned metrics (effect-size fidelity, rank correlation, pathway recovery), not just loss

Results Overview

scGPT was evaluated against state-of-the-art methods on five tasks. In nearly every metric, the fine-tuned scGPT outperforms all baselines — demonstrating genuine knowledge transfer from pre-training.

Key finding:

Pre-training on 10M diverse cells provides a meaningful performance boost over training from scratch, especially on rare cell types and out-of-distribution perturbations.

Task	Outperforms
Batch correction	scVI, Seurat, Harmony
Cell annotation	TOSICA
Perturbation prediction	GEARS, MLP
Multi-omic (paired)	scGLUE, Seurat v4
Multi-omic (mosaic)	scMoMat
GRN inference	Reactome pathway validation

Key Contributions

First Single-Cell Foundation Model

Pre-training on 10M cells with a custom generative objective — establishing a new paradigm for the field.

Novel Attention Masking

Auto-regressive generation for non-sequential data: a unified training scheme for gene-prompt and cell-prompt generation.

Flexible Condition Tokens

Modality, batch, and perturbation state can be injected without retraining the core model — seamless multi-omic extension.

SOTA Across 5+ Tasks

Consistent gains over specialized models in batch correction, annotation, perturbation, multi-omic integration, and GRN inference.

Article | Open access | Published: 16 April 2025

scPRINT: pre-training on 50 million cells allows robust gene network predictions

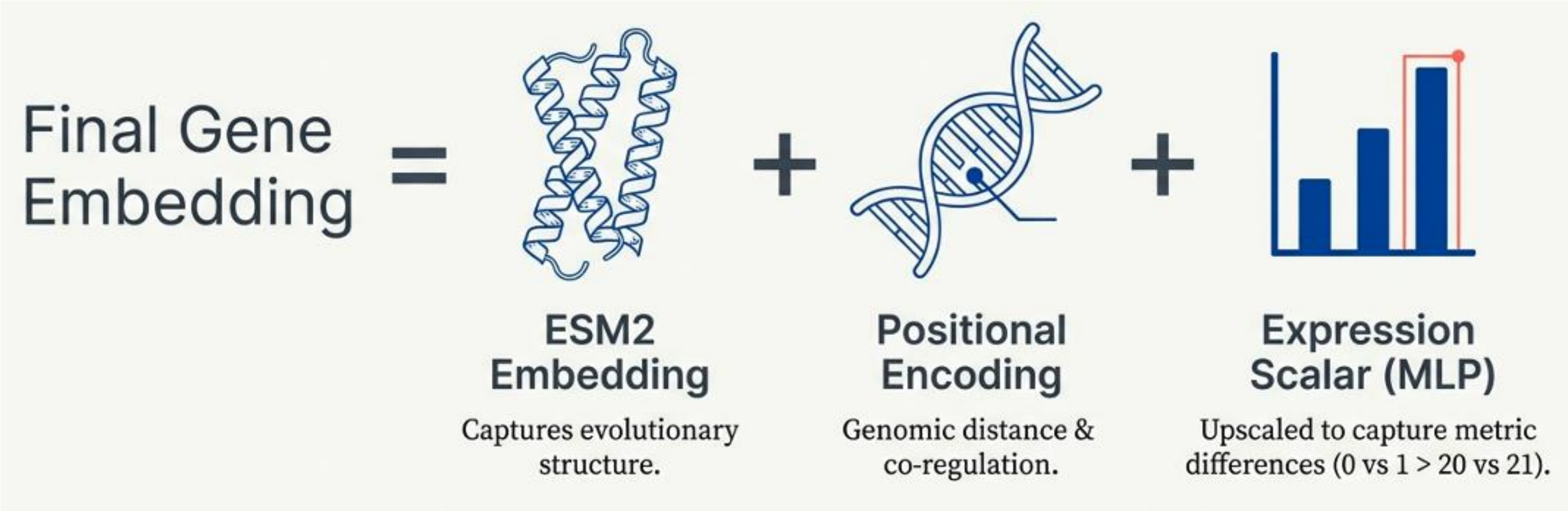
Jérémie Kalfon, Jules Samaran, Gabriel Peyré & Laura Cantini

Nature Communications 16, Article number: 3607 (2025) | Cite this article

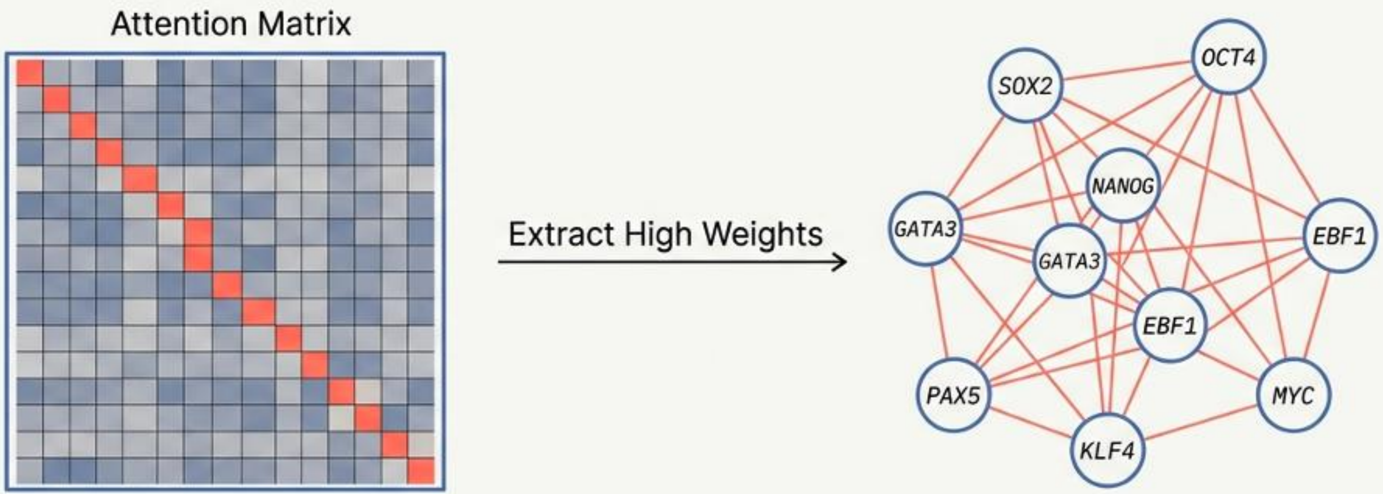
Abstract

A cell is governed by the interaction of myriads of macromolecules. Inferring such a network of interactions has remained an elusive milestone in cellular biology. Building on recent advances in large foundation models and their ability to learn without supervision, we present scPRINT, a large cell model for the inference of gene networks pre-trained on more than 50 million cells from the cellxgene database. Using innovative pretraining tasks and model architecture, scPRINT pushes large transformer models towards more interpretability and usability when uncovering the complex biology of the cell. Based on our atlas-level benchmarks, scPRINT demonstrates superior performance in gene network inference to the state of the art, as well as competitive zero-shot abilities in denoising, batch effect correction, and cell label prediction. On an atlas of benign prostatic hyperplasia, scPRINT highlights the profound connections between ion exchange, senescence, and chronic inflammation.

a foundation model designed for
Gene Network inference.

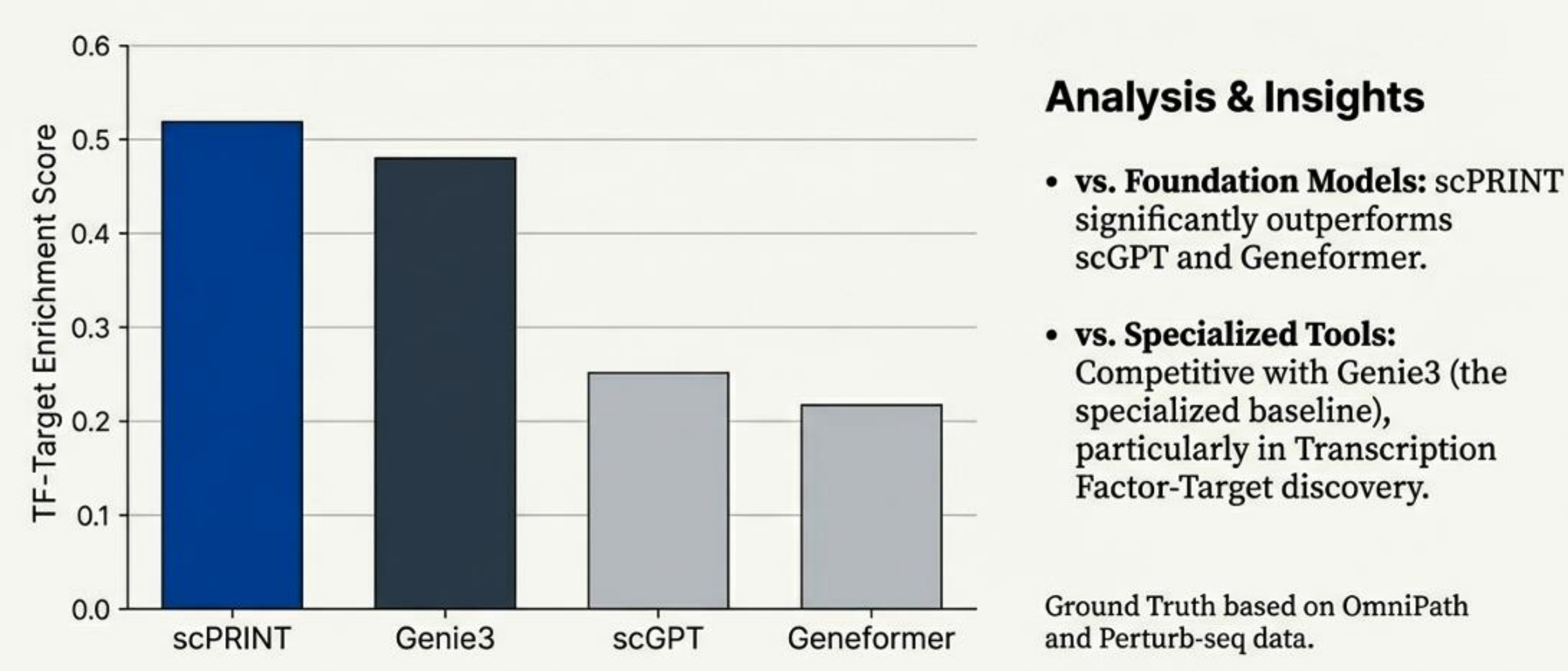


Zero-shot Inference: Attention Map as Gene regulatory Networks



Hypothesis: Attention weights ≈ Molecular interactions.
Capability: Generates cell-type specific networks (unlike static databases).
Scale: Inference on up to 20,000 genes via Flash Attention 2.

a foundation model designed for
Gene Network inference.



5

Virtual Cell Challenge

Toward a Turing Test for the Virtual Cell

Roohani, Hua, Tung et al. · Arc Institute · Cell, June 2025

What Is a "Virtual Cell"?

> A **virtual cell** is a computational model that learns the relationship between cell state and function, and predicts the consequences of perturbations—such as gene knockdowns or drug applications—across cell types and contexts.

Why does this matter?

- Understanding how cells respond to perturbations is central to basic biology and drug discovery
- Experimental testing of all perturbations × all cell types is **impossibly expensive**
- A model that can generalize across contexts could revolutionize biology and medicine

Why Do We Need a Benchmark for Virtual Cells?

Why Now? Conditions Have Changed

- Massively improved single-cell measurement technologies (scRNA-seq, CRISPRi screens)
- Richer perturbational datasets — genome-wide Perturb-seq at scale
- Advances in machine learning and foundation models (scGPT, GEARS)

But no standardized benchmarks exist to evaluate whether models capture generalizable biological structure vs. dataset-specific patterns.

Inspiration: CASP for Protein Folding

- CASP (1994–) created a recurring blind competition for protein structure prediction
- It catalyzed the field and ultimately enabled AlphaFold2 — which solved protein folding
- ImageNet catalyzed computer vision in the same way
- **The Virtual Cell Challenge aims to do the same for predictive cell biology**

Past Competitions and Remaining Gaps

Competition	What It Tested	Gap
Broad Cancer Immunotherapy Challenge	Broad phenotypic shifts in T cells	No gene expression resolution
NeurIPS-Kaggle 2023	Gene expression response to **small molecules**	Limited to chemicals; no genetic perturbations
Virtual Cell Challenge (2025)	Gene expression response to **genetic perturbations**	← *This work*

Why genetic perturbations?

- Precisely defined targets → ideal for probing causal gene-function relationships
- Transcriptional changes can be more subtle → harder to predict → better test

Task: Context Generalization

The key challenge: generalization across cell types

Task formulation:

- Participants have seen perturbation effects in **multiple other cell contexts**
- A **subset** of perturbation effects in H1 hESCs is provided (few-shot adaptation)
- Goal: predict the effects of **held-out perturbations** in H1 hESCs

Why few-shot, not zero-shot?

> Most published datasets span only a handful of cell lines — true zero-shot generalization to new cell states is likely premature at this stage.

Task Design: core components & Three Metrics

Task: Predict Gene Expression Response to CRISPRi Perturbations in H1 hESCs

Core components of the challenge:

- Annual, open competition hosted by Arc Institute
- Purpose-built high-quality benchmark dataset (H1 hESC line)
- Public leaderboard for real-time progress tracking
- Three complementary evaluation metrics
- Training data from the Arc Virtual Cell Atlas + public datasets

Website: virtualcellchallenge.org

The Three Evaluation Metrics

Combined score weighted across all three metrics. Minimum thresholds enforced on all metrics — prevents gaming by models that excel at one dimension at the expense of others.

Metric 1
Differential Expression Score
How accurately does the model predict DE genes?

Metric 2
Perturbation Discrimination Score
Can the model distinguish between perturbations?

Metric 3
Mean Absolute Error (MAE)
Global accuracy across all genes

Benchmark Dataset: High-Quality CRISPRi Screen

Dataset Generation Pipeline

- Start: Broad low-depth screen of 2,500 CRISPRi perturbations in H1 hESCs
- Select 300 perturbations covering broad range of effect sizes (strong → subtle → negligible) and diversity of phenotypic responses (clustering-based selection)
- High-depth targeted sequencing using 10x Genomics Flex chemistry (fixed-cell, reduced batch effects)

Competition Dataset Split

Training set

150 perturbations · ~150,000 cells · Released at launch

Validation set

50 perturbations · ~50,000 cells · Live leaderboard

Test set

100 perturbations · ~100,000 cells · Released 1 week before deadline · Final rankings

Additional Training Resources: Arc Virtual Cell Atlas — scBaseCount + Tahoe-100M (>100M cells, largest public perturbational scRNA-seq dataset). Combined: >350 million cells and counting.

The Benchmark in Context

What good performance would demonstrate:

- A model can generalize gene expression predictions to a new, unseen cell type (H1 hESC)
- Using only a **few-shot subset** of H1 hESC perturbations for adaptation
- Achieving balanced scores on all three complementary metrics
- Reproducing biologically meaningful differential expression signatures

Key Quality Metrics Achieved

Parameter	Value
Total cells profiled	~300,000
Perturbations profiled	300 genes (CRISPRi)
Median cells / perturbation	~1,000 cells
Mean reads per cell	~93,000
Median gene UMIs per cell	~50,000

Goals of the Virtual Cell Challenge

The challenge is designed to achieve three outcomes:

1. Surface best practices and data considerations

- Training set quality, sequencing depth, platform choice
- Guidelines for reproducible and principled data generation

2. Identify methodological bottlenecks

- Clarify which model and data choices drive generalization
- Converge on reproducible standards for perturbation modeling

3. Highlight the importance of data quality

- Cell coverage, sequencing depth, and technology choice
- By underscoring these issues, drive higher-quality community data generation

Future Directions

Expanding the challenge over time:

Near-term expansions:

- Combinatorial perturbations (gene × gene, gene × drug)
- Cross-cell-type generalization at scale
- Multi-modal predictions (proteomics, epigenomics)

Longer-term vision:

- Temporal and spatial dimensions
- Multicellular systems
- **Inverse design:** identify perturbations that achieve a desired cellular effect (therapeutic relevance)

> 🔄Metrics and datasets will evolve each year based on insights from prior challenge results — this is a **living benchmark**, not a static test.

The Broader Vision: A Turing Test for the Virtual Cell

> "Virtual cells are poised to become foundational tools for biology, and to ensure that they reach their potential, we need clear and rigorous evaluations."

Analogies:

- **CASP** → catalyzed protein structure prediction → AlphaFold
- **ImageNet** → catalyzed computer vision
- **Virtual Cell Challenge** → could catalyze predictive cell biology

The ultimate goal:

> A model indistinguishable from experiment — a true Turing test for the virtual cell

Summary

Component	Key Detail
Task	Predict gene expression response to genetic perturbations in H1 hESCs (context generalization)
Format	Few-shot: subset of H1 hESC perturbations provided; 100 held-out test perturbations
Dataset	300 CRISPRi perturbations × ~1,000 cells each; 10x Genomics Flex; high depth
Metrics	DE score + perturbation discrimination score + MAE (combined + minimum thresholds)
Training data	Arc Virtual Cell Atlas (>350M cells) + public perturbation datasets
Goal	Reproducible, fair benchmarking to surface generalizable models of cell behavior

Participate: virtualcellchallenge.org | **Data:** arcinstitute.org/tools/virtualcellatlas

Appendix: Key Definitions

CRISPRi (CRISPR interference): A tool to repress (silence) gene expression using a catalytically dead Cas9 fused to a transcriptional repressor, guided to a specific genomic locus by a guide RNA.

scFG (single-cell functional genomics): Coupling genetic perturbations with single-cell transcriptomics to measure how gene expression changes in response to each perturbation at single-cell resolution.

Perturb-seq: A widely used scFG approach combining pooled CRISPR perturbations with scRNA-seq profiling.

UMI (Unique Molecular Identifier): Molecular barcodes used in scRNA-seq to count individual RNA molecules, reducing PCR amplification bias.

Flex chemistry: 10x Genomics' fixed-cell scRNA-seq protocol that allows multiplexed sample pooling and reduces batch effects through cell fixation before library preparation.

For Predictive Biology

THE GOAL

To build 'Virtual Cells'—AI models capable of predicting cellular responses to perturbations indistinguishably from

THE BOTTLENECK

Progress is stalled by a lack of standardized evaluations and high-quality, reproducible ground-truth data.

THE INITIATIVE

The Virtual Cell Challenge (VCC)—an annual, open-source competition launching in 2025 to evaluate model generalization.

THE ASSET

Powered by a purpose-built, high-depth H1 hESC dataset generated specifically for this challenge by the Arc Institute.

Summary: Six Practitioner Takeaways

1. Architecture matters: StripedHyena 2 shows that complementary operator specialization beats redundant operator design. For byte-level and genomic sequences, multi-hybrid architectures are the right tool.

2. Genomic LMs can predict clinical variant effects zero-shot: Evo 2 achieves state-of-the-art on noncoding SNVs and indels — classes where protein-specific models cannot produce predictions at all.

3. Long context + base-pair resolution + multimodal learning are simultaneously achievable: AlphaGenome dissolves what appeared to be fundamental trade-offs through careful systems design.

4. Separate cell state from intervention effect: scGPT's state-conditioned perturbation modeling consistently outperforms models that conflate baseline identity with causal effect across out-of-distribution cell types.

5. Closed-loop discovery accelerates beyond passive scaling: LUMI-Lab found genuinely novel lipid chemistry in 10 iterations that passive training on historical data would not have found.

6. Benchmarks are infrastructure: The Virtual Cell Challenge provides the ground truth machinery needed to distinguish genuine progress from overfitting to training distributions.